

Jessica Bean

jessicabbean@gmail.com | (206) 591-0312 | linkedin.com/in/jessicabbean | jessicabriandebbean.github.io

PROFESSIONAL SUMMARY

Data Scientist with 10+ years of experience developing and deploying machine learning models for prediction, recommendation, and anomaly detection in large-scale enterprise environments. Strong foundation in deep learning architectures including transformers and neural embeddings, with hands-on experience in NLP, classification, and forecasting systems. Skilled in designing and analyzing A/B tests to measure model impact, building automated data pipelines on cloud platforms, and communicating data-driven insights to cross-functional stakeholders. Passionate about applying rigorous analytical methods to improve digital customer experiences.

TECHNICAL SKILLS

Programming: Python, R, SQL, Git, Visual Studio Code

Machine Learning: Transformer architectures, neural networks, gradient boosting, collaborative filtering, NLP, classification, clustering, embeddings, fine-tuning pre-trained models, attention mechanisms

Deep Learning Frameworks: PyTorch, TensorFlow, scikit-learn, spaCy, Hugging Face

Data and Visualization: Pandas, NumPy, Power BI, Tableau, Matplotlib, Plotly

Cloud and Infrastructure: AWS (SageMaker, S3, EC2), Azure, Docker, automated data pipelines

Experimentation: A/B testing design and analysis, causal inference, Bayesian methods, statistical significance testing

Other: Agile/Scrum, cross-functional collaboration, technical documentation, code review

PROFESSIONAL EXPERIENCE

Senior Data Scientist

Naval Undersea Warfare Center | Keyport, WA | October 2023 – October 2025

- Design, train, and evaluate deep learning models including transformer-based architectures for product lifecycle prediction and anomaly detection, improving long-term forecasting accuracy across multiple product lines.
- Develop and deploy NLP classification and entity extraction pipelines using spaCy and Hugging Face transformers to automate document tagging and semantic search across large unstructured datasets.
- Architect and maintain automated data pipelines on Azure cloud infrastructure, enabling distributed analytics and secure data sharing across multiple business units using a data mesh framework.
- Lead A/B testing and post-launch analysis for model iterations, diagnosing model performance and contributing to data-informed product feature improvements that delivered measurable cost savings.
- Collaborate with cross-functional teams of engineers, product managers, and analysts to prototype, train, test, and monitor ML models in production-serving environments.
- Mentor data science teams from University of Washington and University of Tennessee on machine learning models and stochastic optimization projects, resulting in \$10M in lifecycle cost savings.
- Author strategic white papers on supply chain resilience leveraging AI, big data, and retrieval-augmented generation (RAG) approaches for predictive and conversational analytics applications.

Data Scientist

Naval Undersea Warfare Center | Keyport, WA | September 2020 - October 2023

- Developed patent-pending machine learning models combining bathtub curve distributions with Gaussian methods for 1-5 year obsolescence prediction, now deployed in enterprise recommendation systems.
- Built interactive dashboards and reporting tools (Power BI, Tableau) for real-time risk scoring, supply chain forecasting, and personalized recommendations across multiple product lines.

- Designed and implemented DART (Decision Analytics Reporting Tool), a decision-support platform with recommendation and ranking features adopted organization-wide by technical and non-technical users.
- Directed data-driven inventory reclamation and recommendation initiatives resulting in \$100M in cost avoidance through intelligent matching and collaborative filtering of obsolete components.
- Extracted and prepared large datasets using SQL, Hive, and Spark for analysis; built and maintained data pipelines supporting model training, evaluation, and monitoring workflows.
- Supported model deployment workflows including prototyping, testing, and monitoring, while developing best practices for code review, documentation, and long-term maintainability.

Operations and Data Analyst

Naval Undersea Warfare Center | Keyport, WA | May 2015 - September 2020

- Performed exploratory data analysis and statistical modeling on complex operational datasets, translating findings into actionable business insights for executive stakeholders.
- Developed automated reporting and visualization solutions for KPI tracking, enabling data-informed iteration on operational processes and resource allocation.
- Managed cross-functional collaboration with engineering, product, and operations teams to scope data science projects from intake through deployment and reporting.
- Presented quarterly analytical findings and model performance metrics to technical and non-technical audiences, driving adoption of data-driven decision-making.

EDUCATION

PhD, Information Science (Data Science Focus) - Naval Postgraduate School - In Progress

Research: Qualitative and Quantitative Methods, Network Science, Knowledge Transfer

Master of Science, Data Analytics - Pennsylvania State University - December 2020

Bachelor of Arts, Business Analytics and Economics - California State University, Fullerton - May 2015

SELECTED PRESENTATIONS AND PUBLICATIONS

- Technology Expo Presenter: Mammal Distribution and Environmental Factors in Puget Sound using ML (2025)
- Technology Expo Presenter: Forecasting Obsolescence Models using Deep Learning (2020-2025)
- Poster Presenter: AI and ML Symposium on NLP, Forecasting, and Game Theory (2019-2023)
- Strategic white paper on AI-driven supply chain resilience and predictive analytics